FINAL EXAM DEFINITIONS and THEOREMS

ELEMENTARY OPERATIONS:

$\mathcal{E}_{ij} : V^m \to V^m$; $\mathcal{E}_{i\alpha}$, $\mathcal{E}_{ij\alpha} \to$ automorphisms.

$\mathcal{E}_{ij}(v_1, \dots, v_i, \dots, v_j, \dots, v_m) = (v_1, \dots, v_j, \dots, v_i, \dots, v_m)$

$\mathcal{E}_{i\alpha}(v_1, \dots, v_i, \dots, v_m) = (v_1, \dots, \alpha v_i, \dots, v_m)$

$\mathcal{E}_{ij\alpha}(v_1, \dots, v_i, \dots, v_j, \dots, v_m) = (v_1, \dots, v_i + \alpha v_j, \dots, v_j, \dots, v_m)$

EQUIVALENT LISTS OF VECTORS:

$X \sim X' \Leftrightarrow \exists \varphi : V^m \to V^m$ s.t. $\varphi(X) = X'$ or $\varphi(X') = X$

($\varphi \to$ equivalence relation (homo + ref + sym + tranz.)

ECHELON FORM of a matrix:

0 vectors under the "main diag."

RANK of a MATRIX:

$\text{rank}(A) = \dim \langle a_1, \dots, a_m \rangle$

$= \dim \langle a^1, \dots, a^m \rangle$

$= \text{rank}(C) \to$ echelon form of A

$= r \to$ nr. of non-zero rows

INVERSE of a MATRIX: $\exists A^{-1} \Leftrightarrow \det(A) \neq 0$.

$[A \,\vdots\, I_m] \to [I_m \,\vdots\, A^{-1}]$; OR: $A^{-1} = \dfrac{1}{\det A} \cdot A^*$

KRONECKER-CAPPELI:

$\text{rank}(A) = \text{rank}(\bar{A}) \Rightarrow (S)$ comp.

ROUCHÉ: (comp. nedet, det = 0)

$\mu_c = 0$ (all) $\Rightarrow (S)$ comp.

CRAMER: (comp. det, det $\neq 0$)

$x_i = \dfrac{\Delta x_i}{\Delta} \leftarrow$ det. with column of free terms

$\Rightarrow$ unique solution.

GAUSS: (rows!)

$\bar{A} \to$ echelon form $\to$ replace it with the system of $x, y, z, \dots$ using those coordinates.

GAUSS-JORDAN (rows!)

$\bar{A} \to$ identity form $\to$ direct results.

THEOREM:

$[f(v)]_{B'} = [f]_{BB'} \cdot [v]_B$

$[f(v)]_{B'}$ → coordinate column of the image

$[f]_{BB'}$ → matrix of a linear map

$[v]_B$ → column matrix.

MATRIX of a VECTOR:

$B = (v_1, \dots, v_m)$ basis of V.

$v = k_1 v_1 + \dots + k_m v_m$

$\Rightarrow [v]_B = \begin{bmatrix} k_1 \\ \vdots \\ k_m \end{bmatrix}$

THEOREM:

$\text{rank}(f) = \text{rank}([f]_{BB'})$

f linear map

$f : V \to V'$

B, B' bases of V, V'.

rank of a linear map = rank of the matrix of that linear map.

ELEMENTARY MATRICES:

$E_{ij}, E_{i\alpha}, E_{ij\alpha}$ obtained from applying $\mathcal{E}_{ij}, \mathcal{E}_{i\alpha}, \mathcal{E}_{ij\alpha}$ on I_m.

(list of column vectors)

MATRIX = "row of columns"

$(a^1, a^2, \dots, a^m) = A \in \mathcal{M}_{m,m}(K)$

$$\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} \dots \begin{pmatrix} a_{1m} \\ a_{2m} \\ \vdots \\ a_{mm} \end{pmatrix}$$

(list of row vectors)

MATRIX = "column of rows" ✓

$(a_1, a_2, \dots, a_m) = A \in \mathcal{M}_{m,m}(K)$

$(a_{11}, a_{12}, \dots, a_{1m}) \quad (a_{m1}, a_{m2}, \dots, a_{mm})$

MATRIX OF THE LIST OF VECTORS:

$B = (v_1, \dots, v_m)$ basis of V.

$X = (u_1, \dots, u_m)$ list of vectors

$$\begin{cases} u_1 = a_{11} v_1 + \dots + a_{1m} v_m \\ \vdots \\ u_m = a_{m1} v_1 + \dots + a_{mm} v_m \end{cases}$$

$\Rightarrow [X]_B = \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & & \\ a_{m1} & \dots & a_{mm} \end{bmatrix}$

ON ROWS

DIMENSION, BASIS of a LIST OF Vect

$\dim \langle X \rangle = \text{rank}(A)$

a basis of $\langle X \rangle \to$ the non-zero vectors of $C \sim A$.

$\boxed{\dim(S+T) = \dim S + \dim T - \dim(S \cap T)}$

MATRIX OF THE LINEAR MAP:

$B = (v_1, \dots, v_m)$ basis of V.

$B' = (v_1', \dots, v_m')$ basis of V'.

$f : V \to V'$. (if $V = V' \Rightarrow [f]_B = [f]_{BB}$)

$$\begin{cases} f(v_1) = a_{11} \cdot v_1' + \dots + a_{m1} \cdot v_m' \\ f(v_m) = a_{1m} \cdot v_1' + \dots + a_{mm} \cdot v_m' \end{cases}$$

$\Rightarrow [f]_{BB'} = \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & & \vdots \\ a_{m1} & & a_{mm} \end{bmatrix}$

ON COLUMNS

THEOREM:

$f, g \in Hom_K(V, V')$

$h \in Hom_K(V', V'')$

$k \in K$

$[f+g]_{BB'} = [f]_{BB'} + [g]_{BB'}$

$[kf]_{BB'} = k \cdot [f]_{BB'}$

$[h \circ f]_{BB''} = [h]_{B'B''} \cdot [f]_{BB'}$

FIRST DIMENSION THEOREM:

$\dim(\mathrm{Ker}\, A) + \dim(\mathrm{Im}\, A) = m.$

TRANSITION MATRIX (CHANGE MATRIX):

$B = (v_1, \dots, v_m)$

$B' = (v_1', \dots, v_m')$

$$\begin{cases} v_1' = t_{11} v_1 + \dots + t_{m1} v_m \\ \vdots \\ v_m' = t_{1m} v_1 + \dots + t_{mm} v_m \end{cases}$$

$$\Rightarrow T_{BB'} = \begin{bmatrix} t_{11} & \dots & t_{1m} \\ \vdots & & \\ t_{m1} & \dots & t_{mm} \end{bmatrix} = [1_V]_{B'B}$$

THEOREM:

$$\boxed{T_{BB''} = T_{BB'} \cdot T_{B'B''}}$$

$B = (v_1, \dots, v_m)$

$B' = (v_1', \dots, v_m')$

$B'' = (v_1'', \dots, v_m'')$

THEOREM:

$T_{BB'}$ invertible with the inverse $T_{B'B}$

B, B' bases of V.

THEOREM:

$$\boxed{[v]_B = T_{BB'} \cdot [v]_{B'}}$$

B, B' bases of V.

THEOREM:

$f \in Hom_K(V, V')$

$$\boxed{[f]_{B_2 B_2'} = T^{-1}_{B_1' B_2'} \cdot [f]_{B_1 B_1'} \cdot T_{B_1 B_2}}$$

B_1, B_2 bases of V

B_1', B_2' bases of V'

EIGENVALUES:

$p(\lambda) = 0$

$\Rightarrow$ solutions...

EIGENVECTORS:

$$A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = O_m$$

$\Rightarrow x_1, \dots, x_m.$

ISOMORPHISM OF VECTOR SPACES: (THEOREM)

$\varphi : Hom_K(V, V') \to \mathcal{M}_{m,m}(K)$ linear map

$\varphi(f) = [f]_{BB'}$

ISOMORPHISM OF VECTOR SPACES + RINGS: (THEOREM)

$\varphi : End_K(V) \to \mathcal{M}_m(K)$ linear map.

$\varphi(f) = [f]_B$

COROLLARY

$f \in End_K(V)$

$f \in Aut_K(V) \Leftrightarrow \det([f]_B) \neq 0$

EIGENVECTOR, EIGENVALUE DEFINITION:

$f \in End_K(V)$

$v \in V$ eigenvector of f if $\exists \lambda \in K$ s.t.
$\neq 0$

$\boxed{f(v) = \lambda \cdot v}$; λ = eigenvalue
v = eigenvector

THE SET OF EIGENVECTORS and THE ZEROVE[CTOR]

$V(\lambda) = \{v \in V \mid f(v) = \lambda v\} \to$ Subspace of V.

EIGENSPACE

$V(\lambda)$ + f is endomorphism.

THEOREM:

$[f]_B = A = (a_{ij}) \in \mathcal{M}_m(K)$

f endomorphism.

$\lambda \in K$ eigenvalue of $f \Leftrightarrow \boxed{\det(A - \lambda I_m) = 0}$

THEOREM:

B, B' bases of V.

f endomorphism

$[f]_B = A$; $[f]_{B'} = A'$

$p_A(\lambda) = p_{A'}(\lambda) = \det(A - \lambda I_m)$
$= \det(A' - \lambda I_m)$

CHARACTERISTIC POLYNOMIAL

$p(\lambda) = \det([f]_E - \lambda I_m) = \det(A).$
$= (\lambda - \lambda_1)(\lambda - \lambda_2)$
$= \lambda^2 - \lambda \cdot Tr(A) + \det(A)$

THEOREM:

$f \in End_K(V)$

$$\boxed{[f]_{B'} = [f]_{B'B'} = T^{-1}_{BB'} \cdot [f]_B \cdot T_{BB'}}$$

B, B' bases of V $\boxed{= T_{B'B} [f]_B \cdot T_{BB'}}$

THEOREM:

$A \in \mathcal{M}_m(K)$

$\lambda_1, \dots, \lambda_m$ eigenvalues

$\boxed{\lambda_1 + \dots + \lambda_m = Tr(A)}$

$\boxed{\lambda_1 \cdot \dots \cdot \lambda_m = det(A)}$

DIAGONALIZABLE ENDOMORPHISM:

f endomorphism.

f diag $\Leftrightarrow$ f has m lin. indep. eigenvectors

ex: $\begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & \vdots & & \\ 0 & 0 & & \lambda_m \end{pmatrix}, \lambda_1 \neq \dots \neq \lambda_m = [f]_B$.

KERNEL:

$[f]_E \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ \vdots \end{bmatrix} \in Ker f$

$\left[\begin{array}{c|c} [f]_E & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \end{array}\right] \to$ echelon form $\to \begin{cases} \dots = 0 \\ \dots = 0 \end{cases}$

IMAGE:

$v' \in Im f \Leftrightarrow \exists v$ s.t. $f(v) = v'$.

$Im f$ = lin. ind. vectors in the echelon form of $[f]_E$.

$\dim(Im f) = rank([f]_E)$
$= $ max. nr. of lin. ind. vectors in $[f]_E$.
$= \dim(f(e))$
$= \dim(\langle f(e_1), f(e_2), \dots \rangle)$

$\boxed{f \text{ automorphism} \Leftrightarrow det([f]_E) \neq 0}$

ALWAYS TRUE FOR LINEAR MAPS:

$[\alpha f]_B = \alpha [f]_B$

$[f+g]_B = [f]_B + [g]_B$

$[f \circ g]_B = [f]_B \cdot [g]_B$

(3,2)-parity check code:

2 digits message (k)
3 digits code (m)

rule: $\underbrace{\_\ \_}_{\oplus\ \%2}\ \_$

CAYLEY-HAMILTON:

Every $A \in \mathcal{M}_m(K)$ is a root of its char. polyn.

COROLLARY:

$A \in \mathcal{M}_2(K)$

$\boxed{P_A(\lambda) = \lambda^2 - Tr(A) \cdot \lambda + det(A)}$

$\boxed{A^2 - Tr(A) \cdot A + det(A) \cdot I_2 = O_2}$

(m, k) - code

m = nr. of digits of the code word.
k = nr. of digits of the message

$\boxed{\text{nr. of messages} = 2^k}$

GENERATOR MATRIX

$G = \begin{bmatrix} P \\ \hline I_k \end{bmatrix} = [\gamma]_{EE'}$

$\gamma : \mathbb{Z}_2^k \to \mathbb{Z}_2^m$
(E) (E')

PARITY-CHECK-MATRIX:

$H = [\, I_{m-k} \,\vdots\, P \,]$

CHECK CODE WORDS:

$H \cdot [u] = 0 \Rightarrow [u]$ code word.

MINIMUM HAMMING DISTANCE

d = how many columns added together in H can give me the vector 0 (column) (minimum nr.)

CODE CAN $\boxed{\text{DETECT}}$ t ERRORS:

$t + 1 \leq d$

CODE CAN $\boxed{\text{CORRECT}}$ t ERRORS:

$2t + 1 \leq d$

(3,1) - repeating code:

1 digit message (k)
3 digits code (m)

rule: _ 1 _ _

• „Which codes contain detectable errors?" / „Decode".
↳ check digits with the rule.

• „f, g code words generated by p?"
↳ check $\boxed{\text{degree}(f) < m}$ → yes ⇒ f might be a code word.
→ no ⇒ STOP.
↳ check $\boxed{f : p}$ → yes (residue = 0) ⇒ f is a code word.
→ no (residue ≠ 0) ⇒ f not a code word.

• „generate all the words".
ex: $p = 1 + x^2 + x^3$
(6,3) - code. ⇒ 2^3 messages ⇒ $M = \{000, 001, 010, \dots\}$
<u>011</u>: $m = 0 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 = x + x^2$

$m \cdot x^{m-k} = (x + x^2) \cdot x^{6-3} = x^5 + x^4$

$r = m \cdot x^{m-k} \pmod p$

code word $= v = m \cdot x^{m-k} + r$

• „all code words"; we know $G = \begin{bmatrix} P \\ I_m \end{bmatrix}$, P.
↳ H = ...
↳ $H \cdot [u] = 0 \Rightarrow [u] = \dots$ = the format for all code words.

• „encode ... using the generator matrix G."

$G = \begin{bmatrix} P \\ I_m \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & \vdots \end{bmatrix}$
↓ ... ↓
$\gamma(e_1)$ ... $\gamma(e_m)$

$1101 = e_1 + e_2 + e_4 \mid \gamma$
$\Rightarrow \gamma(1101) = \gamma(e_1 + e_2 + e_4) = \gamma(e_1) + \gamma(e_2) + \gamma($

• „(m, k) - code generated by p..."
$\gamma : \mathbb{Z}_2^k \to \mathbb{Z}_2^m$
$E = (e_1, \dots, e_k)$
$E' = (e_1', \dots, e_m')$
$e_1 = \dots \Rightarrow m$
$m \cdot x^{m-k}$
$r = m \cdot x^{m-k} \bmod p$
$v = r + m \cdot x^{m-k} = \gamma(e_1)$
⋮
$e_k = \dots \Rightarrow \gamma(e_k)$

$\Rightarrow G = \begin{bmatrix} \vdots & & \vdots \end{bmatrix} = \begin{bmatrix} P \\ I_k \end{bmatrix}$
↓ ↓
$\gamma(e_1)$ $\gamma(e_k)$

P...
H..